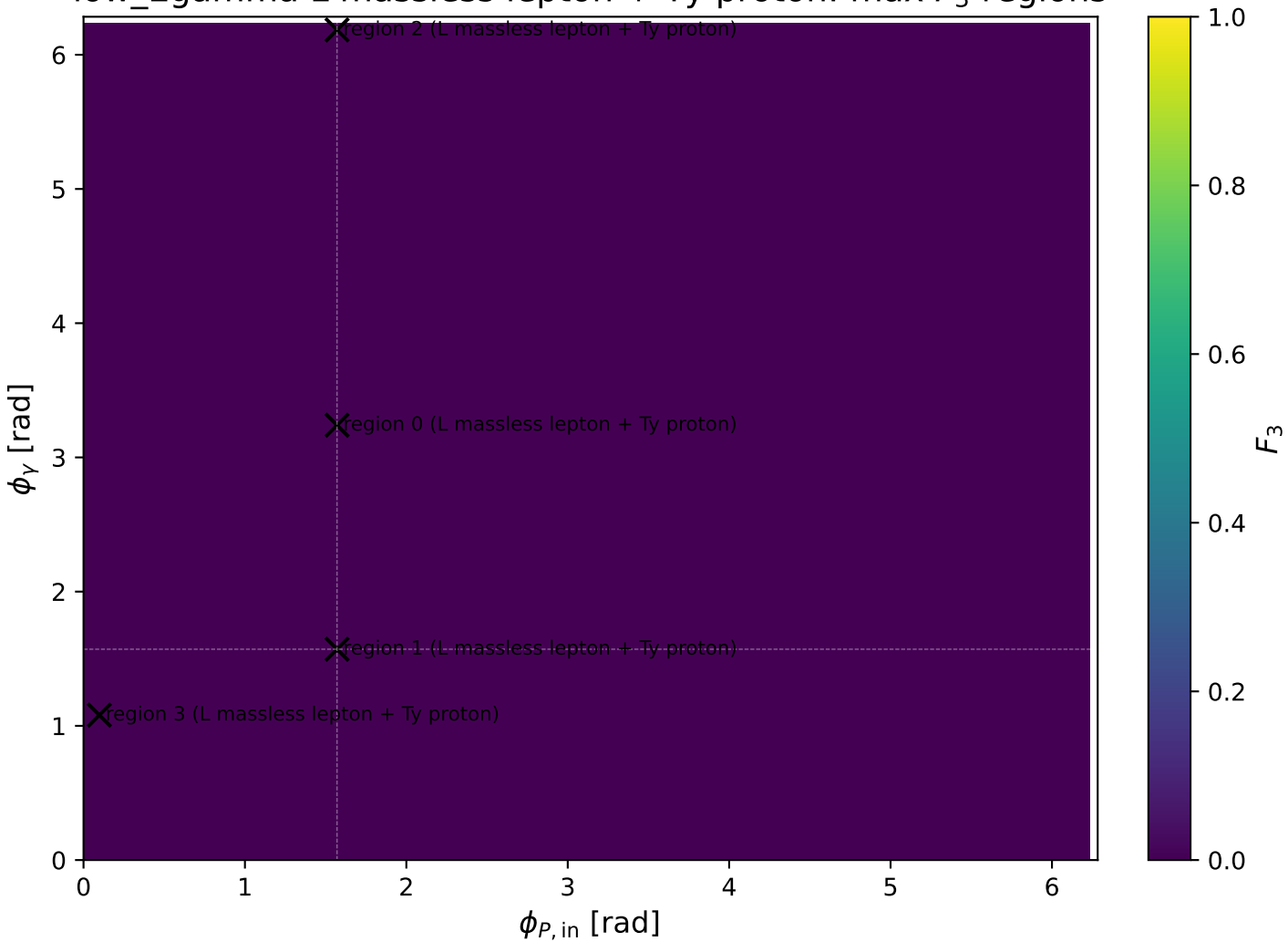
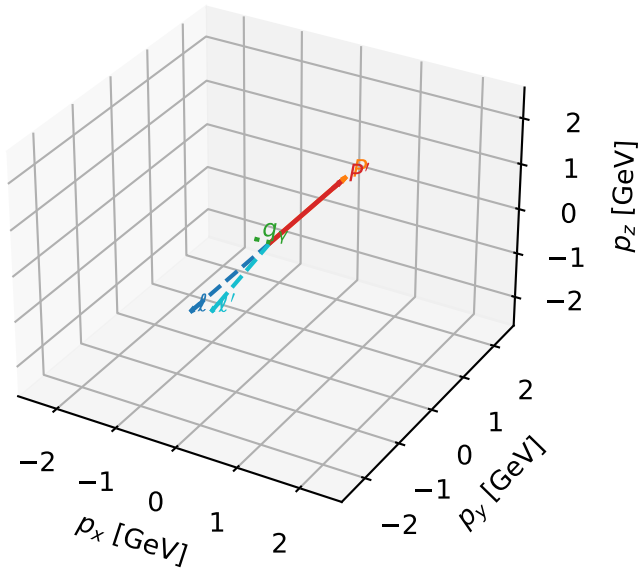


low_Egamma L massless lepton + Ty proton: max F_3 regions



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta

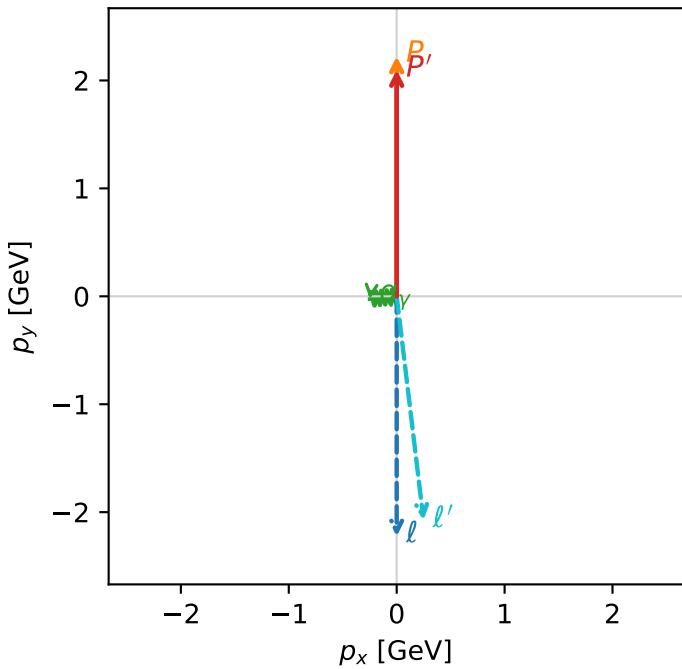


f3_low_egamma_lty_region_0 F_3=2.93827e-08
 region: low_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_{l_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.09905$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_P=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=3.23977$
 $Q^2=0.0661343$, $x_B=0.0501566$, $t=-0.00249164$, $W^2=2.13227$, $y=0.063896$
 $\theta(l', \gamma)=91.2137$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 l' $E= 2.0894$ $p_x= 0.2488$ $p_y= -2.0745$ $p_z= 0.0000$
 P' $E= 2.2991$ $p_x= 0.0000$ $p_y= 2.0991$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2488$ $p_y= -0.0245$ $p_z= 0.0000$

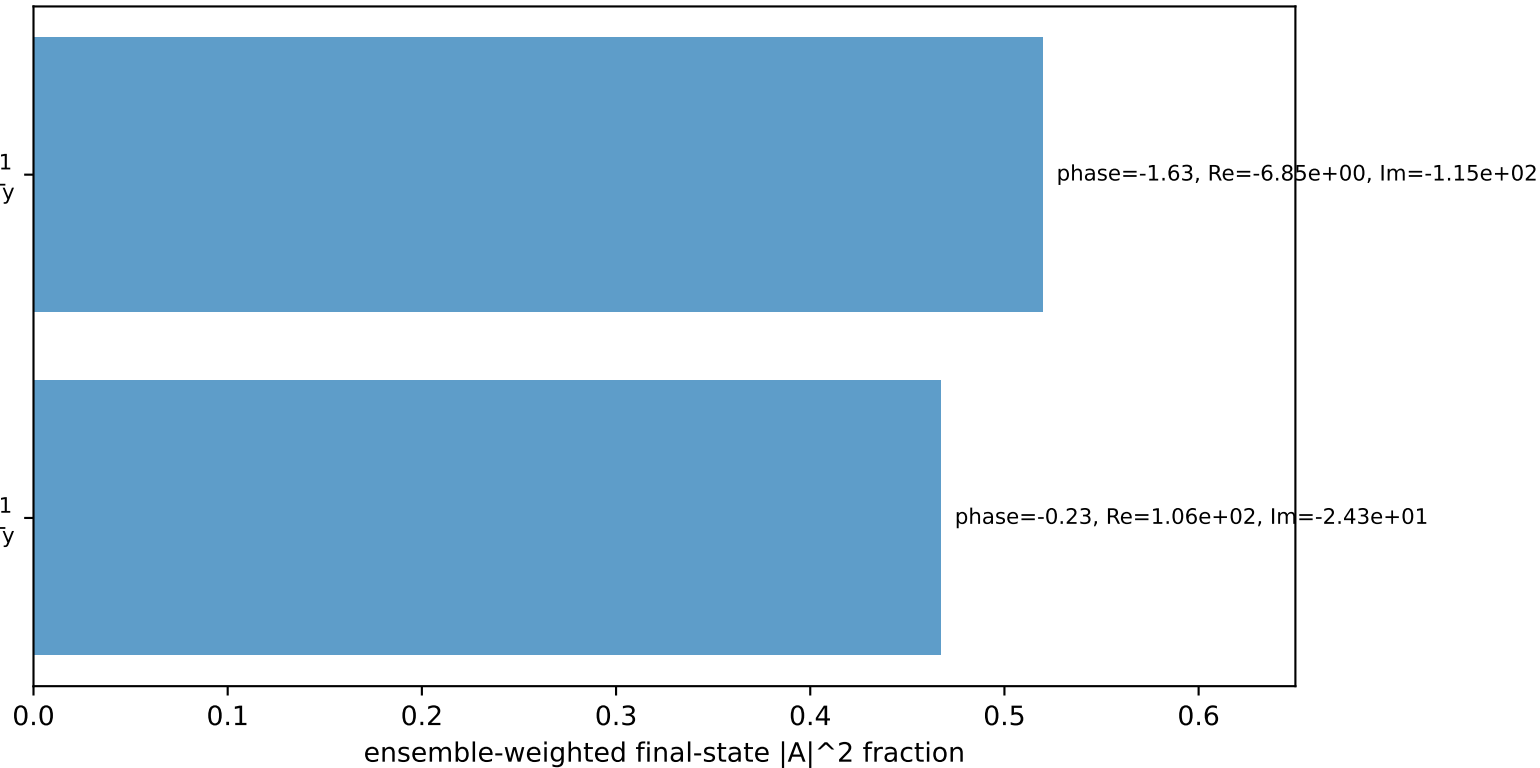
Transverse plane



$h_e=+1, h_p=-1, h_\gamma=+1$
 massless lepton L, proton Ty

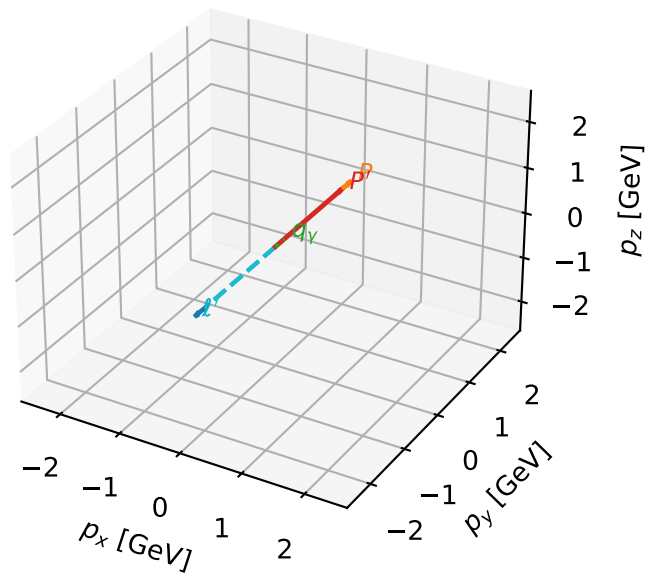
$h_e=+1, h_p=+1, h_\gamma=-1$
 massless lepton L, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=98.7%)



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta

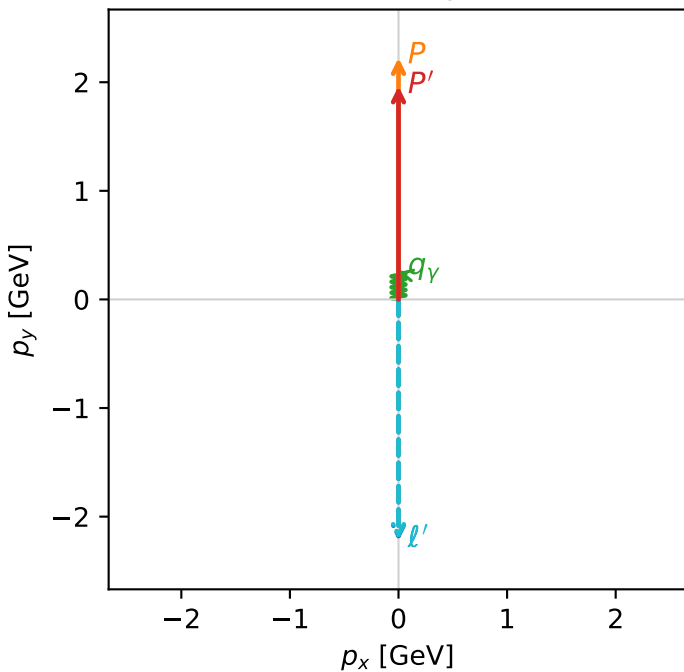


f3_low_egamma_lty_region_1 $F_3=2.43332\text{e-}08$
 region: low_Egamma_LTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.96296$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_P=1.5708$
 $E_Y=0.25$, $\phi_Y=1.5708$
 $Q^2=8.66557\text{e-}12$, $x_B=8.15204\text{e-}11$, $t=-0.0114583$, $W^2=0.986143$, $y=0.00515116$
 $\theta(\ell', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.2130$ $p_x= -0.0000$ $p_y= -2.2130$ $p_z= 0.0000$
 P' $E= 2.1756$ $p_x= 0.0000$ $p_y= 1.9630$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.0000$ $p_y= 0.2500$ $p_z= 0.0000$

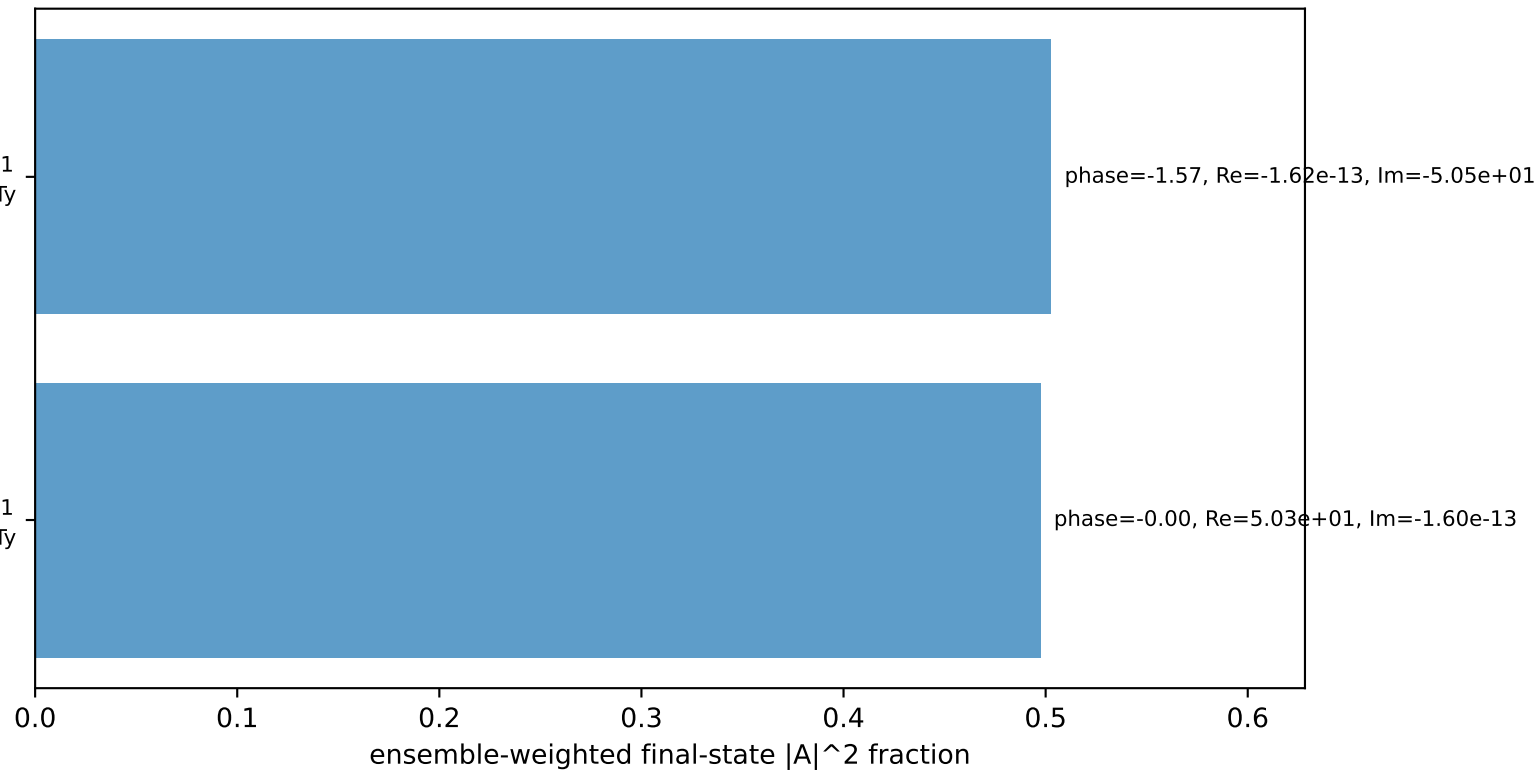
Transverse plane



$h_e=+1$, $h_p=-1$, $h_Y=+1$
 massless lepton L, proton Ty

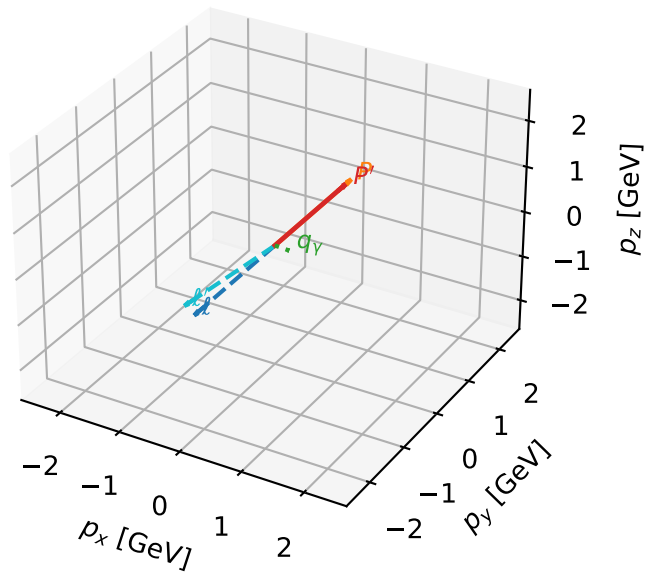
$h_e=+1$, $h_p=+1$, $h_Y=-1$
 massless lepton L, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta

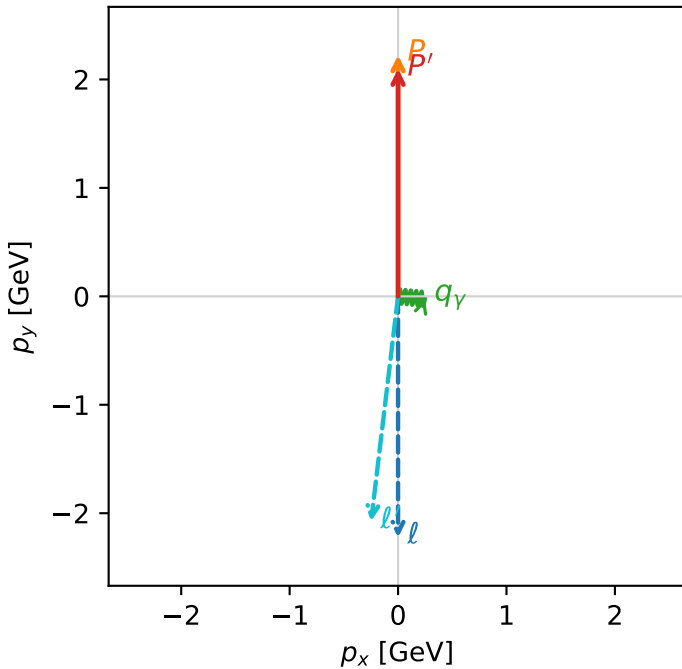


f3_low_egamma_lty_region_2 F_3=2.39909e-08
 region: low_Egamma_LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_{l_0}\rangle \otimes |T_{\gamma,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.09905$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_p=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=6.18501$
 $Q^2=0.0661343$, $x_B=0.0501566$, $t=-0.00249164$, $W^2=2.13227$, $y=0.063896$
 $\theta(l', \gamma)=91.2137$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.0894$ $p_x= -0.2488$ $p_y= -2.0745$ $p_z= 0.0000$
 P' $E= 2.2991$ $p_x= 0.0000$ $p_y= 2.0991$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2488$ $p_y= -0.0245$ $p_z= 0.0000$

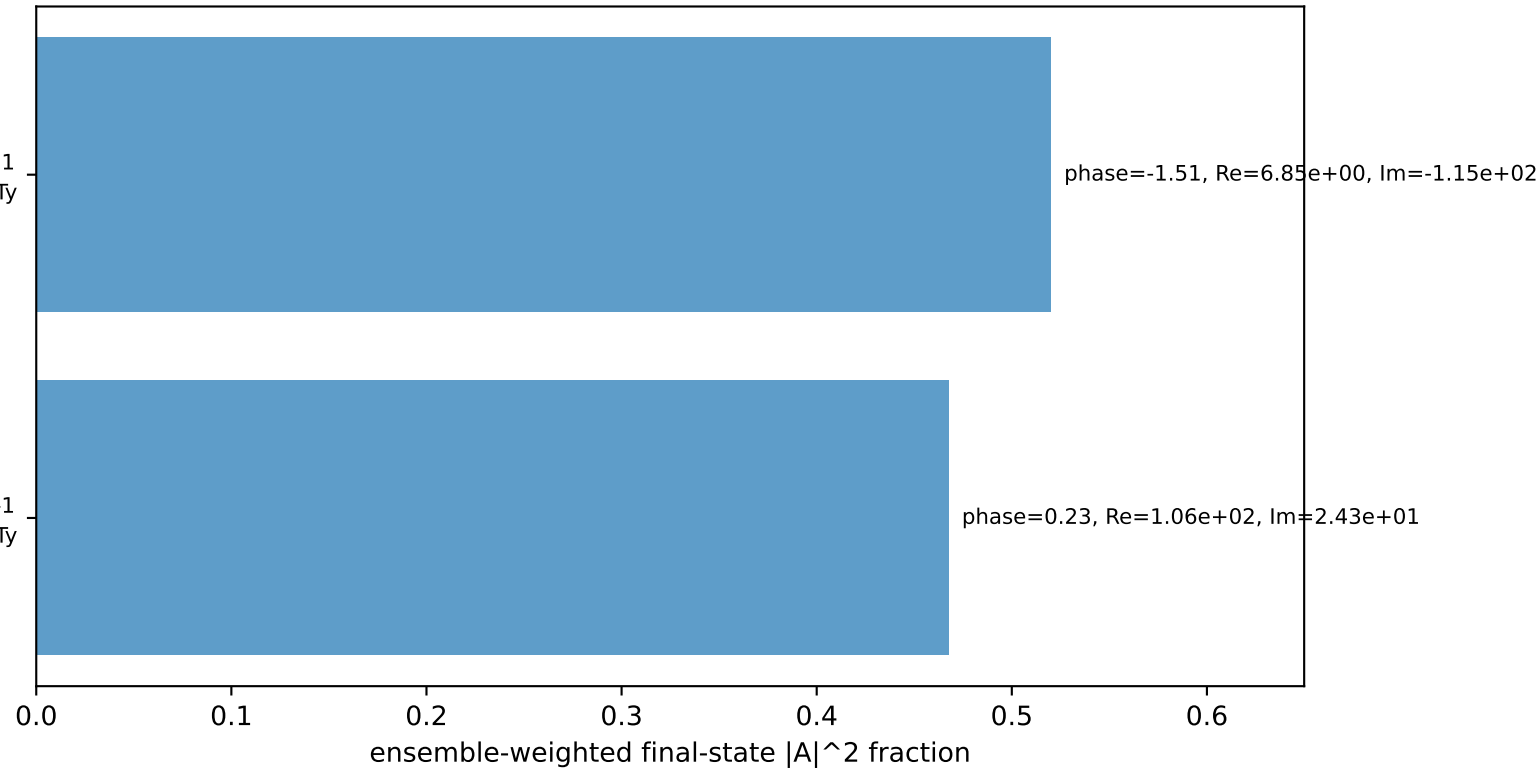
Transverse plane



$h_e=+1, h_p=-1, h_\gamma=+1$
 massless lepton L, proton Ty

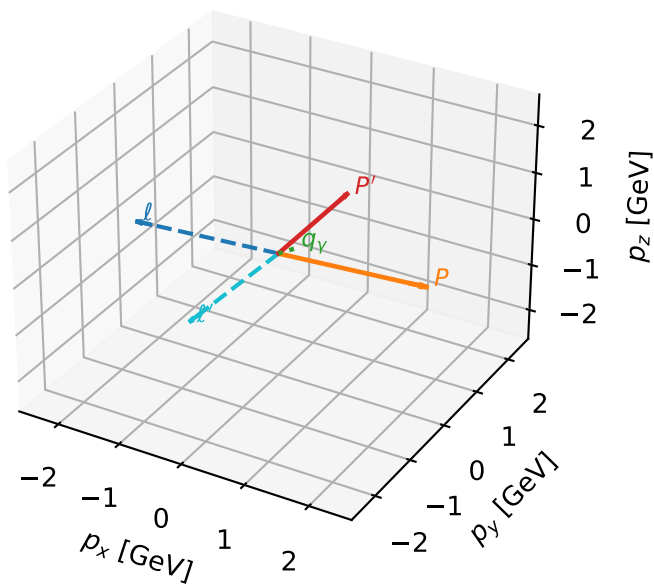
$h_e=+1, h_p=+1, h_\gamma=-1$
 massless lepton L, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=98.7%)



L massless lepton + Ty proton characteristic max F_3 configuration

3D momenta

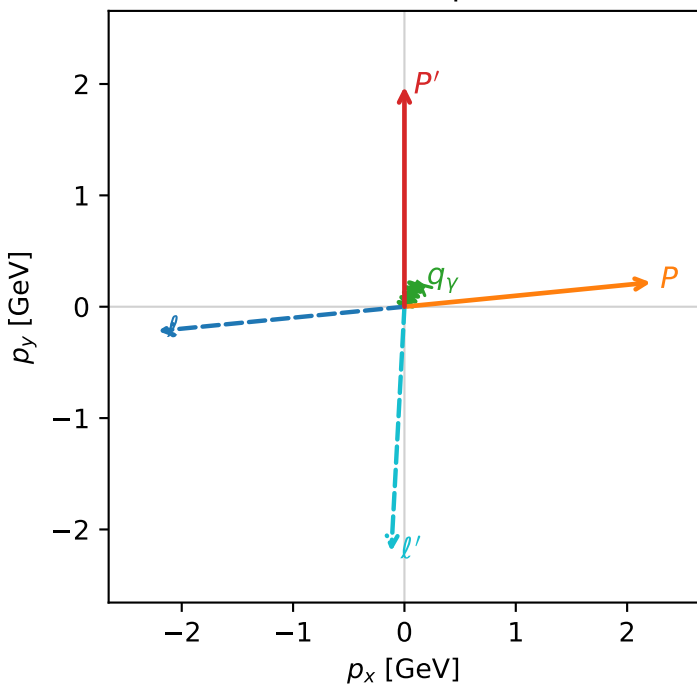


f3_low_egamma_lty_region_3 F_3=5.04352e-09
 region: low_Egamma_LTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.97681$
 $\theta_{in}=1.57079$, $\phi_l=3.23977$, $\phi_P=0.0981748$
 $E_Y=0.25$, $\phi_Y=1.07992$
 $Q^2=8.30952$, $x_B=0.97394$, $t=-7.94272$, $W^2=1.10218$, $y=0.413445$
 $\theta(l', \gamma)=154.945$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -2.2137$ $p_y= -0.2180$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 2.2137$ $p_y= 0.2180$ $p_z= 0.0000$
 l' $E= 2.2005$ $p_x= -0.1178$ $p_y= -2.1973$ $p_z= 0.0000$
 P' $E= 2.1881$ $p_x= 0.0000$ $p_y= 1.9768$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.1178$ $p_y= 0.2205$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)

